小车运行机构设计计算书
Calculation Sheet for Design of Trolley Operation Mechanism

*机构级别：M8
*Mechanism class: M8

T6L3

小车重量：TL=10.8t
Trolley weight: TL=10.8t

小车基距：Bx= 2400mm
Trolley base: Bx=2,400mm

小车轨距：Ly= 2800mm
Trolley track gauge: Ly=2,800mm

小车重心高度（相对轨面）：h_{gt1}= 11000mm（满载，相对于轨顶面）
Trolley's height of center of gravity (relative track surface): h_{gt1}=11,000mm (full load, relative to the track top)

h_{gt2}= 11000mm

小车满载运行速度：v_{y1}= 55m/min
Travelling speed of trolley at full load: v_{y1}=55m/min

0.92m/s

小车空载运行速度：v_{y2}= 55m/min
Travelling speed of trolley at no-load: v_{y2}=55m/min

0.92m/s

许用制动时间：$[t_z]$= 4.0s
Allowable braking time: $[t_z]$=4.0s

许用起动时间：$[t_q]$= 4.0s
Allowable starting time: $[t_q]$=4.0s

许用起、制动加速度：$[a_z]$= 0.23m/s^2
Allowable starting and braking acceleration: $[a_z]$=0.23m/s^2

小车车轮数：n_{lt}= 4
Number of trolley wheels: n_{lt}=4

1.车轮和轨道
1. Wheel and Track

1.1.选型
1.1 Model Selection

选择车轮直径：D_{yt}= 315mm
Diameter of selected wheel:Dyt=315mm

车轮内孔直径：d_{lt}= 100mm
Inner hole diameter of wheel:dlt=100mm

车轮宽度：B=135mm
Wheel width: B=135mm

车轮型号： ZQCL315
Wheel model: ZQCL315

车轮材料：42CrMo
Wheel material: 42CrMo

σ_b= 1080MPa（最小抗拉强度）
σ_b=1080MPa (minimum tensile strength)

单个车轮装配重量：m_{lt}= 50kg
Individual wheel assembly weight: m_{lt}=50kg

车轮轴材料：35CrMo
Wheel axle material: 35CrMo

车轮轴直径：d_{lt}= 100mm
Wheel axle diameter: d_{lt}=100mm

小车轨道型号：方钢
Trolley track model: square steel

轨道尺寸：h1= 50mm
Track size: h1=50mm

b1= 70mm

k= 70mm

r1= 0mm

轨道材料： C45 钢
Track material: C45 steel

σ_b= 560MPa（最小抗拉强度）
σ_b=560MPa (minimum tensile strength)

1.2.踏面疲劳强度校核
1.2 Fatigue Strength Check of Tread

小车自重引起的轮压：P_{Lt11}= 2.70kN（满载）
Wheel load caused by the self-weight of trolley: P_{Lt11}=2.70kN (full load)

P_{Lt12}=2.70kN（空载）
P_{Lt12}=2.70kN (no-load)

起升载荷引起的轮压：P_{Lt1l}= 44.15kN（满载）
Wheel load caused by lifting load: P_{Lt1l}=44.15kN (full load)

P_{Lt12}= 19.50kN（空载）
P_{Lt12}=19.50kN (no-load)

由惯性载荷引起的轮压：P_{Lt31}= 20.48kN（满载）
Wheel load caused by inertial load: P_{Lt31}=20.48kN (full load)

P_{Lt32}= 11.03 kN（空载）
P_{Lt32}=11.03kN (no-load)

正常工作状态下的最小轮压：P_{Ltmin}= 67.72kN
Minimum wheel load in normal working condition: P_{Ltmin}=67.72kN

正常工作状态下的最大轮压：P_{Ltmax}= 79.23kN
Maximum wheel load in normal working condition: P_{Ltmax}=79.23kN

平均轮压：P_{Lmean}=75.39kN $\left(P_{Lmean}=\frac{P_{L\min}+2P_{L\max}}{3}\right)$

Average wheel load: P_{Lmean}=75.39kN $\left(P_{Lmean}=\frac{P_{L\min}+2P_{L\max}}{3}\right)$

与材料有关的许用线接触应力常数：k_1= 7.20　(轨道σb>510，车轮σb>800)
Allowable line contact stress constant related to the material: k_1=7.20 (track: σb>510, wheel: σb>800)

车轮与轨道有效接触宽度：l= 70.0N/mm^2
Effective contact width between the wheel and the track: l =70.0N/mm^2

车轮转速：n_{yt}= 55.58mm
Wheel rotation speed: n_{yt}=55.58mm

转速系数：C_1= 0.92r/min
Speed coefficient: C_1=0.92r/min

工作级别系数：C_2= 0.80　（根据车轮直径和运行速度）
Working class coefficient: C_2=0.80 (according to the wheel diameter and speed)

校核公式：$P_{Lmean} \le k_1 D_{ytl} C_1 C_2$　（机构级别 M8）

Check formula: $P_{Lmean} \leq k_1 D_{ytl} C_1 C_2$ (mechanism class: M8)

$k_1 D_{yt} l C_1 C_2$=116.85kN 车轮满足疲劳强度要求!
$k_1 D_{yt} l C_1 C_2$=116.85kN The wheels meet the fatigue strength requirements!

静强度校核：$P_{L\max} \leq 1.9 k_1 D_{ytl}$
Static strength check: $P_{L\max} \leq 1.9 k_1 D_{ytl}$

$1.9 k_1 D_{yt} l$ =301.64kN 车轮满足静强度要求!
$1.9 k_1 D_{yt} l$ =301.64kN The wheel meets the static strength requirements!

2.轴承

2. Bearing

调心滚子轴承： 22217 CC/W33
Self-aligning roller bearing: 22217 CC/W33

重量： 2.68 kg
Weight: 2.68kg

轴承支承距离：L_z= 117 mm （车轮中心到轴承中心的距离）
Bearing support distance: L_z=117mm (distance from the wheel center to the bearing center)

轴承外径：D=150 mm
Outer diameter of bearing: D=150mm

轴的直径：d_z= 85 mm
Diameter of axle: d_z=85mm

倒角：r=3.0 mm
Chamfering: r=3.0mm

$$L_{10h} = \frac{10^6}{60 n_{yt}} \left(\frac{f_t \times C_r}{P} \right)^{\varepsilon}, P = f_d (X F_r + Y F_a)$$

C_{r0}= 325 kN

额定寿命：$[L_{10h}]$=12500 小时 （与机构的使用等级有关）T6
Rated service life: $[L_{10h}]$=12,500h (related to the service level of the mechanism) T6

温度系数：f_t =1.00 （工作温度<120°C）
Temperature coefficient: f_t =1.00 (operating temperature: <120°C)

寿命系数：ε=3.33 （滚子轴承）
Service life coefficient: ε=3.33 (roller bearing)

冲击载荷系数：f_d =1.20 （中等冲击力）
Impact load coefficient: f_d =1.20 (medium impact force)

轴向载荷系数：γ= 0.10
Axial load coefficient: γ=0.10

实际轴向载荷：$F_{a\,1}$= 0.00kN
Actual axial load: F_{a1}=0.00kN

$F_{a\,2}$= 6.73kN $F_a = \gamma P_L$

实际径向载荷：$F_{r\,1}$=42.73kN $\left(F_r = \frac{P_L}{2} \pm \frac{F_a}{L_z} \cdot \frac{D_{yc}}{2}\right)$

Actual radial load: F_{r1}=42.73kN $\left(F_r = \frac{P_L}{2} \pm \frac{F_a}{L_z} \cdot \frac{D_{yc}}{2}\right)$

$F_{r\,2}$= 24.60kN

判断系数：e=0.30
Judgment coefficient: e=0.30

径向动载系数：X=1.00 1.00
Radial dynamic load coefficient: X=1.00 1.00

轴向动载系数：Y=2.30 2.30
Axial dynamic load coefficient: Y=2.30 2.30

当量动载荷：P_1= 51.27kN
Equivalent dynamic load: P_1=51.27kN

P_2= 48.10kN

P= 50.22kN

额定动载荷：C_r= 246kN
Rated dynamic load: C_r=246kN

实际轴承寿命 L_{10h}= 59879 小时轴承满足寿命要求!
Actual service life of the bearing: L_{10h}=59,879h. The bearing meets the service life requirements!

3.三合一减速机
3. Three-in-one Reducer

3.1.载荷计算
3.1 Load Calculation

自重载荷：P_{Gt1}= 282.5 kN（满载）
Self-weight load: P_{Gt1}=282.5kN (full load)

P_{Gt2}= 182.5 kN（空载）
P_{Gt2}=182.5kN (no-load)

小车水平惯性载荷：P_{At1}= 6.6 kN（满载）
Horizontal inertial load of the trolley: P_{At1}=6.6kN (full load)

P_{At2}= 4.3kN（空载）
P_{At2}=4.3kN (no-load)

车轮轴承摩擦系数：μ_t= 0.015（滚子轴承）
Friction coefficient of wheel bearing: μ_t=0.015 (roller bearing)

滚动摩擦系数：k_t= 0.30（轨道为钢质、平顶）
Coefficient of rolling friction: k_t=0.30 (the track is made of steel with the flat top)

附加摩擦阻力系数：C_f = 1.30（有轮缘有水平滚轮，分别驱动）
Additional friction resistance coefficient: C_f =1.30 (with wheel rim and horizontal roller, driven respectively)

运行摩擦阻力：
$$P_m = mg\frac{\mu d + 2k}{D_{yt}}C_f$$

Travelling friction resistance:
$$P_m = mg\frac{\mu d + 2k}{D_{yt}}C_f$$

Pmt1= 2.50 kN （满载）
Pmt1=2.50kN (full load)

Pmt2= 1.61 kN （空载）
Pmt2=1.61kN (no-load)

运行坡度阻力：P_P= mg.sinα= mg . m_p
Slope resistance of trolley travelling: P_P=mg.sinα=mg.mp

坡道阻力系数：m_p= 0.002
Slope resistance coefficient: m_p=0.002

P_{pt1}= 0.58kN（满载）
P_{pt1}=0.58kN (full load)

P_{pt2}= 0.37kN（空载）
P_{pt2}=0.37kN (no-load)

小车运行静阻力：$P_j = P_m + P_p$
Static travelling resistance of trolley: $P_j=P_m+P_p$

P_{jt1}= 3.1kN （满载）
P_{jt1}=3.1kN (full load)

P_{jt2}= 2.0kN （空载）
P_{jt2}=2.0kN (no-load)

3.2.电动机选型
3.2 Motor Selection

电动机台数：m=2 （分别驱动）
Number of motors: m=2 (driven respectively)

运行机构总效率：η_{tz} =0.95
Overall efficiency of the operating mechanism: η_{tz}=0.95

机构静功率：$$P_N = \frac{P_{jt} \cdot v_{yt}}{1000\eta_{tz} \cdot m} = 1.48 \quad \text{kW}$$

Static power of the mechanism: $$P_N = \frac{P_{jt} \cdot v_{yt}}{1000\eta_{tz} \cdot m} = 1.48 \quad \text{kW}$$

考虑 1.4 倍增大系数 2.07kW
Considering the enhancement coefficient for 1.4 times: 2.07kW

所选电动机的参数如下：
The parameters of the selected motor are as follows:

型号：**诺德（三合一平行轴减速机）SK3282ABGH** 100 AP/4
Model: **NORD (three-in-one parallel shaft reducer) SK3282ABGH** 100 AP/4

额定功率：P_n= 3kW
Rated power: P_n=3kW

温度系数：f_T= 0.77（50℃)
Temperature coefficient: f_T=0.77 (50℃)

起动系数：f_H= 0.98
Starting coefficient: f_H=0.98

额定转速：n_n= 1450r/min
Rated speed: n_n=1,450r/min

机构的理论传动比：$i_0 = 26.09$（$i_0 = n_n/n_{yt}$）
Theoretical transmission ratio of the mechanism: $i_0=26.09$ ($i_0=n_n/n_{yt}$)

实际传动比：$i = 25.88$
Actual transmission ratio: $i=25.88$

额定转矩：$M_2 = 19.59$N.m
Rated torque: $M_2=19.59$N.m

平均起动转矩倍数：$\lambda_{as} = 1.60$
Average starting torque multiple: $\lambda_{as}=1.60$

最大起动转矩倍数：$\lambda_m = 2.4$
Maximum starting torque multiple: $\lambda_m=2.4$

转动惯量：$J_1 = 0.067$kg.m^2
Rotational inertia: $J_1=0.067$kg.m^2

3.3.实际电动机转速
3.3 Actual Motor Speed

转速校核公式：

$$n_N = \frac{v_{yt} i}{\pi D_{yt}}$$

Speed check formula:

$$n_N = \frac{v_{yt} i}{\pi D_{yt}}$$

电动机实际转速：n_{yc}=1438 m/min （满载）
Actual motor speed: n_{yc}=1,438 m/min (full load)

4.高速轴制动器
4. High-speed Axle Brake

$$M_Z = \frac{1}{m}\left\{\frac{\left(P_{WI} + P_p - P_m\right) D_{yt} \cdot \eta}{2i} + \frac{n \sum J}{9.55 t_z}\right\}$$

电机轴上转动件的转动惯量：$J_2 = 0.423$kg.m^2
Rotational inertia of rotating parts on the motor shaft: $J_2=0.423$kg.m^2

折 算 到 电 机 轴 上 的 总 转 动 惯 量 ： ΣJ = 2.140kg.m^2

$$\sum J' = 1.15m\left(J_1 + J_2\right) + \frac{(Q+TL)D_0^2\eta}{4i^2}$$

Total rotational inertia converted to the motor shaft: ΣJ=2.140kg.m^2

$$\sum J' = 1.15m\left(J_1 + J_2\right) + \frac{(Q+TL)D_0^2\eta}{4i^2}$$

制动器的制动力矩：M_Z= 35.07N.m
Braking torque of the brake: M_Z=35.07N.m

最大制动力矩：M_{Zmax}= 40N.m　　制动力矩 80-100Nm
Maximum braking torque: M_{Zmax}=40N.m　　Braking torque: 80-100Nm

个数：　m_z=2　台
Qty,: m_z=2 sets

制动器选择满足要求!
The selection of the brake meets the requirements!

5.机构起动时间和加速度的验算
5. Checking Calculation for Starting Time and Acceleration of the Mechanism

5.1.满载起动时间和起动平均加速度验算
5.1 Checking Calculation for Starting Time and Average Starting Acceleration at Full Load

①起动时间
① Starting Time

$$t_q = \frac{n\sum J}{9.55\left(mM_{dq} - M_{dj}\right)}$$

电动机起动力矩：M_{dq}= 62.7 N.m
Starting torque of the motor: M_{dq}=62.7N.m

电动机静阻力矩：M_{dj}= 19.7 N.m　$\left(M_{dj} = \dfrac{P_{jt}D_{yt}}{2000i\eta}\right)$

Static resistance torque of the motor: M_{dj}=19.7N.m　$\left(M_{dj} = \dfrac{P_{jt}D_{yt}}{2000i\eta}\right)$

折算到电机轴上的总转动惯量：ΣJ= 2.250 kg.m^2
Total rotational inertia converted to the motor shaft: ΣJ=2.250 kg.m^2

$$\sum J = 1.15m(J_1 + J_2) + \frac{(Q+TL)D_0^2}{4i^2\eta}$$

起动时间：t_q= 3.2 s
Starting time: t_q=3.2s

允许的起动时间： [t_q]= 4.0 s　满足要求!
Allowable starting time: [t_q]=4.0s　Meet the requirements!

②起动平均加速度
② Average Starting Acceleration

平均加速度：a_q= 0.29 m/s^2
Average acceleration: a_q=0.29 m/s^2

允许的起动平均加速度：[a_q]= 0.23m/s^2　满足要求!
Allowable average starting acceleration: [a_q]=0.23m/s^2 Meet the requirements!

6.缓冲器
6. Buffer

6.1.选型
6.1 Model Selection

$$W = \frac{m_{hc}v_c^2}{2} - P_{mt}S = E_t - P_{mt}S = \frac{1}{2}cS^2 \times n$$

最大减加速度：a_{max}=4.0m/s^2
Maximum deceleration and acceleration: a_{max}=4.0m/s^2

运行阻力：P_{mt} = 2.61kN
Operating resistance: P_{mt}=2.61kN

碰撞瞬时动能：E_t =4.5kN.m
Instantaneous collision kinetic energy: E_t=4.5kN.m

Δ=36458.1

缓冲行程：S=210mm
Buffer stroke: S=210mm

碰撞质量：m_{hc} =10.80t
Collision mass: m_{hc}=10.80t

碰撞前的速度折减系数：100%

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Company Stamp:	Date: December 16, 2024	

Velocity reduction coefficient before collision: 100%

碰撞速度：v_c =0.92m/s
Collision velocity: v_c=0.92m/s

单个缓冲器吸收的能量：W=2.0kJ
Energy absorbed by single buffer: W=2.0kJ

小车缓冲器型号：**HCPU80-A**
Model of trolley buffer: **HCPU80-A**

缓冲器容量：[W]=3.615kJ 容量满足要求！
Buffer capacity: [W]=3.615kJ The capacity meets the requirements!

最大撞击力：P_e =280.0kN
Maximum impact force: P_e=280.0kN

缓冲器许用行程：[S]=60mm
Allowable stroke of the buffer: [S]=60mm

弹性系数：c=2.0083kN/mm（弹性缓冲器）
Elastic coefficient: c=2.0083kN/mm (elastic buffer)

缓冲器效率：η_h =0.85
Buffer efficiency: η_h=0.85

单边个数：n=2.0
Unilateral number: n=2.0

6.2.单个缓冲器末端力校核
6.2 Check for End Force of Single Buffer

缓冲器的末端力： $$P_{\max} = \frac{cS}{\eta} = 141.76 \quad \text{kN}$$

End force of the buffer: $$P_{\max} = \frac{cS}{\eta} = 141.76 \quad \text{kN}$$

缓冲力满足要求！
The buffer force meets the requirements!